GITAM Deemed to be University

Mechanical Engineering | Course Code: ME304

Heat & Mass Transfer - Heat & Mass Transfer Thermal Engineering Handout

---

Overview: Transient lump capacitance analysis, forced convection over flat plates, radiation sh

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Steady & Transient Conduction
Fourier law, 3D heat conduction equation in Cartesian and cylindrical coordinates, Critical thicknes

Unit 2: Forced & Natural Convection
Boundary layer approximations, Nusselt, Reynolds, and Prandtl numbers, Empirical correlations for fl

Unit 3: Thermal Radiation
Stefan-Boltzmann law, Planck law, Wien displacement law, Emissivity, View factor algebra, Radiation

Unit 4: Heat Exchangers & Mass Transfer
Overall heat transfer coefficient, Fouling factor, LMTD method, Effectiveness-NTU method, Fick first

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the temperature distribution and heat dissipation from a pin fin of uniform cross-section with
2. Calculate the effectiveness and heat transfer rate of a counter-flow heat exchanger using the Effectiv
3. State reciprocity, summation, and superposition rules for radiation shape factors and find the shape f

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Frank P. Incropera, David P. DeWitt - Fundamentals of Heat and Mass Transfer Wiley 7th Ed
* J.P. Holman - Heat Transfer McGraw-Hill 10th Ed

Verified NoteVault Institutional Archive - Authorized for Registered Student Use Only